

Some simple polygon problems solved with orientations tests

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Rodrigo Silveira

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SOME SIMPLE POLYGON PROBLEMS

Problem 1: Point in polygon test

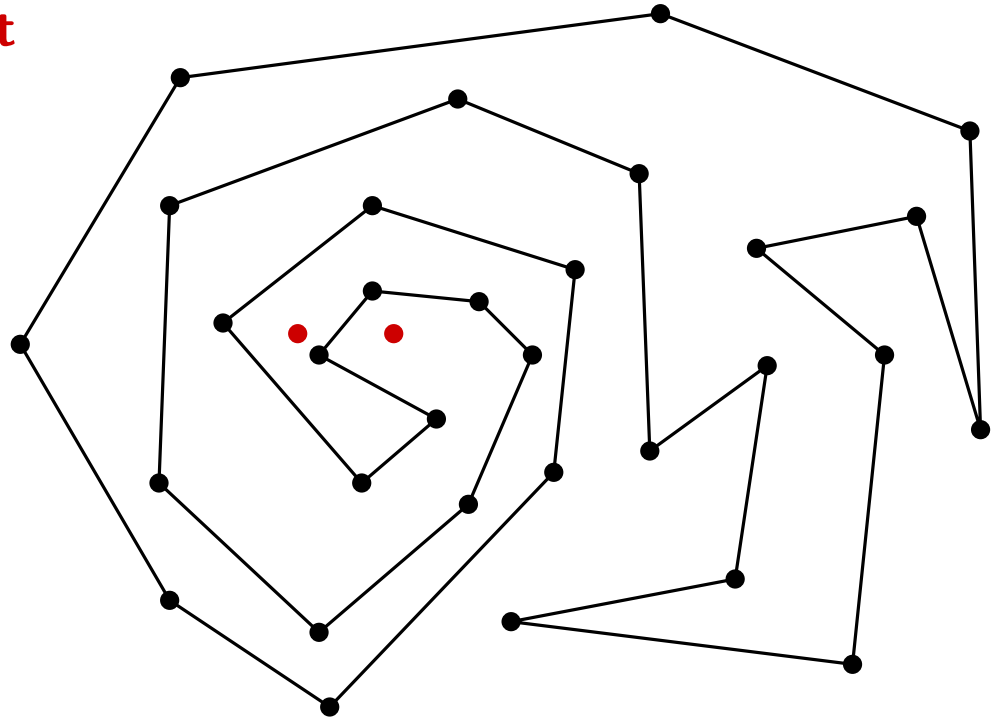
Input:

A polygon $p_1, p_2, \dots, p_n$ (sorted)

A query point q

Output:

Yes/No $q \in P$



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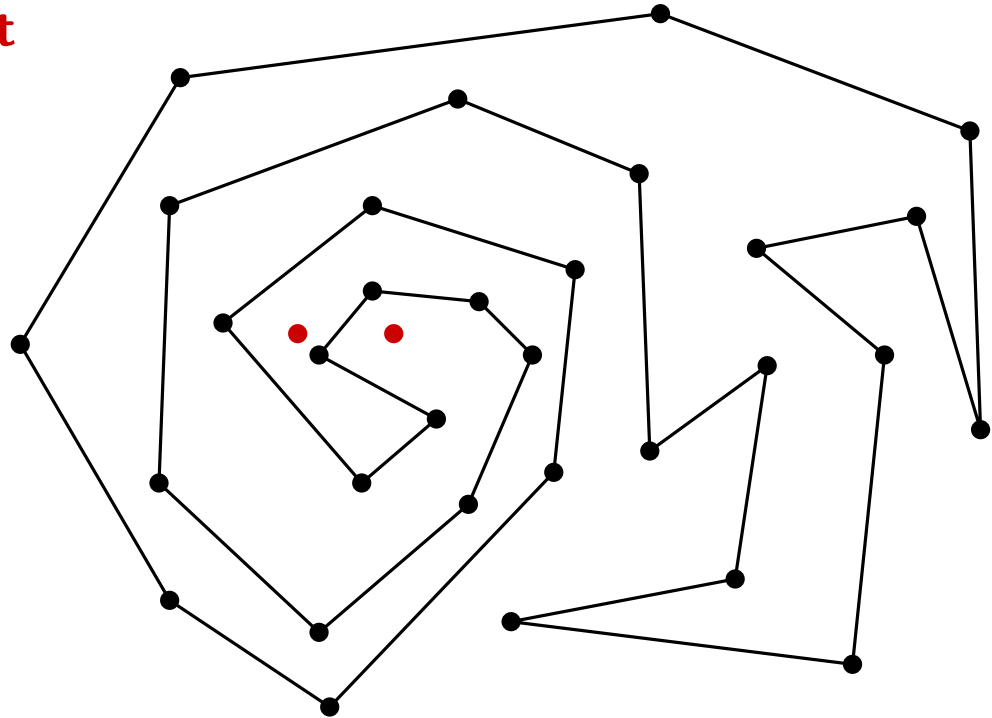
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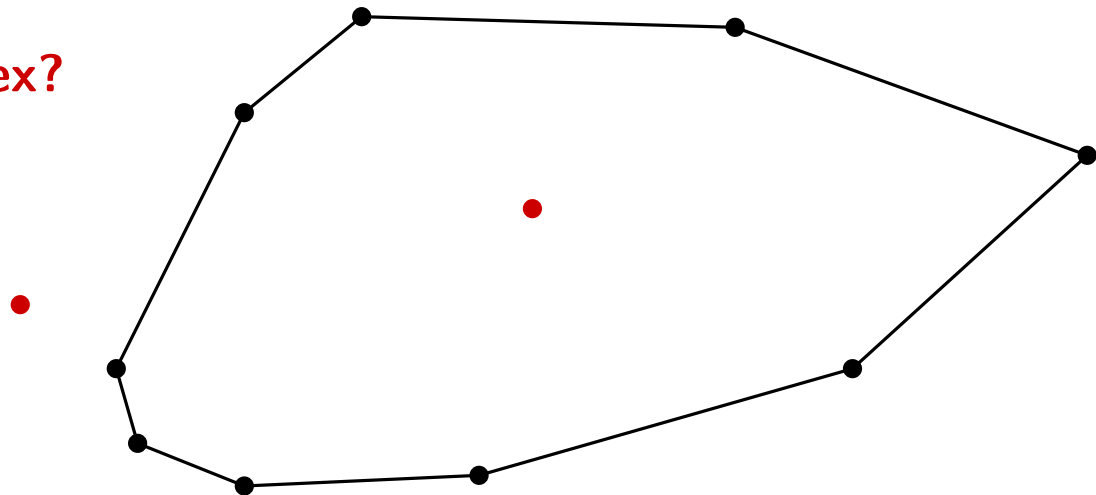
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Yes/No $q \in P$



What if the polygon is convex?



SOME SIMPLE POLYGON PROBLEMS

Problem 2: Intersection test between line and polygon

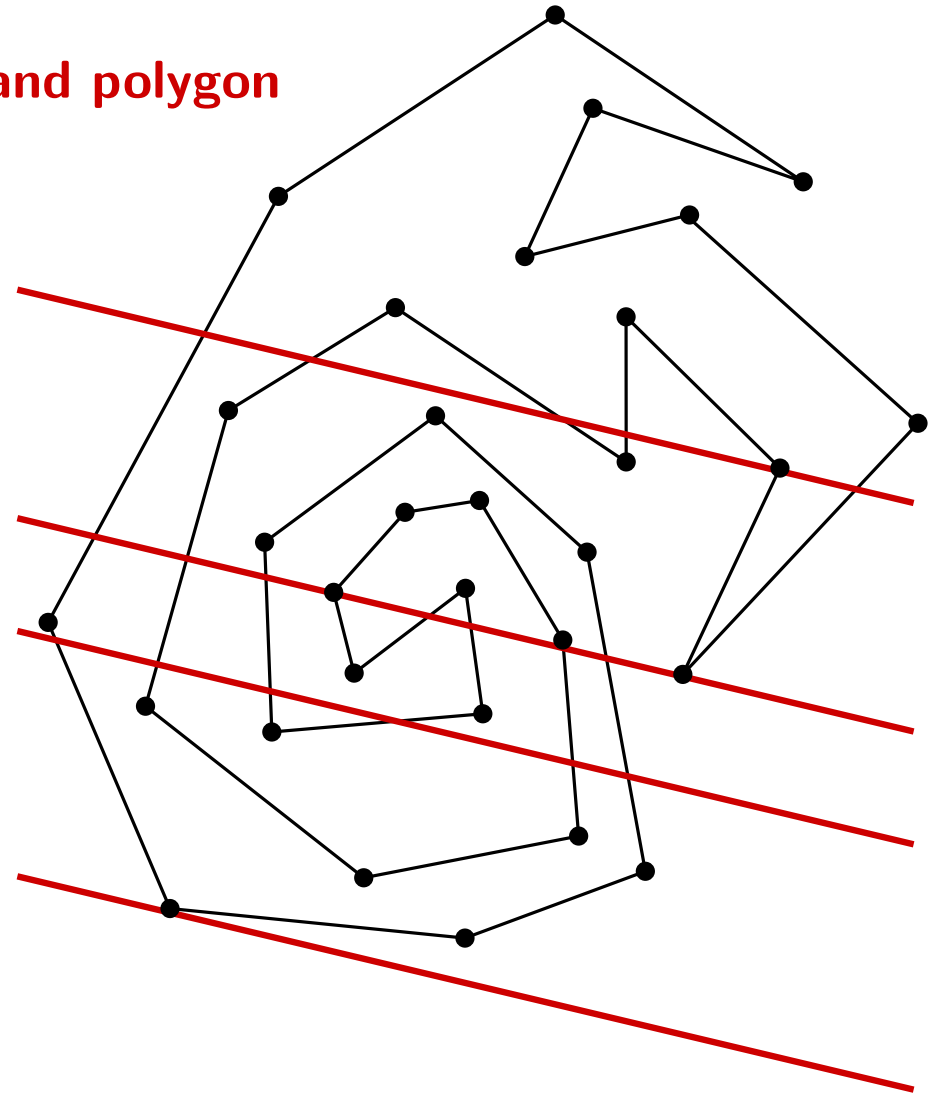
Input:

ℓ : a line (through p and q)

P : a polygon (with sorted vertices $p_1, p_2, \dots, p_m$)

Yes/No they intersect.

If they do, the edges of P intersecting ℓ



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Problem 2: Intersection test between line and polygon

Input:

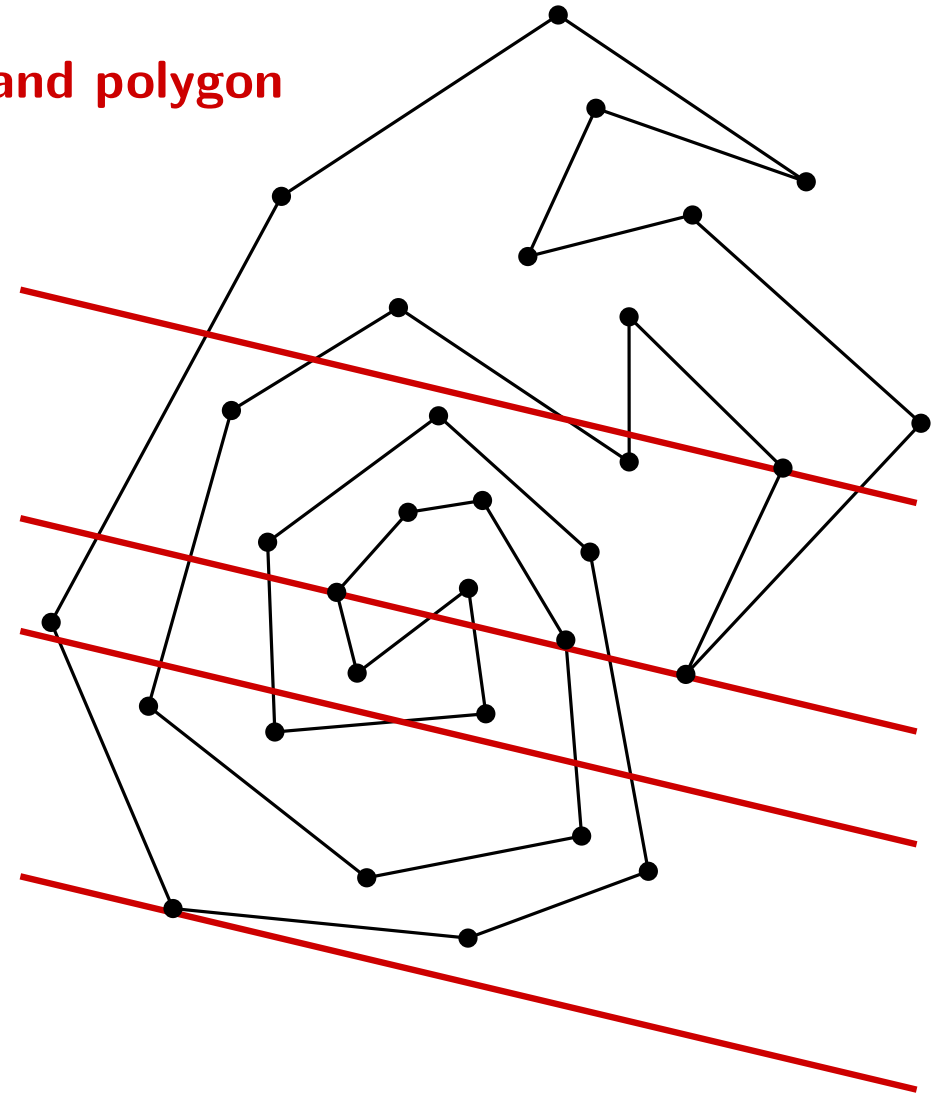
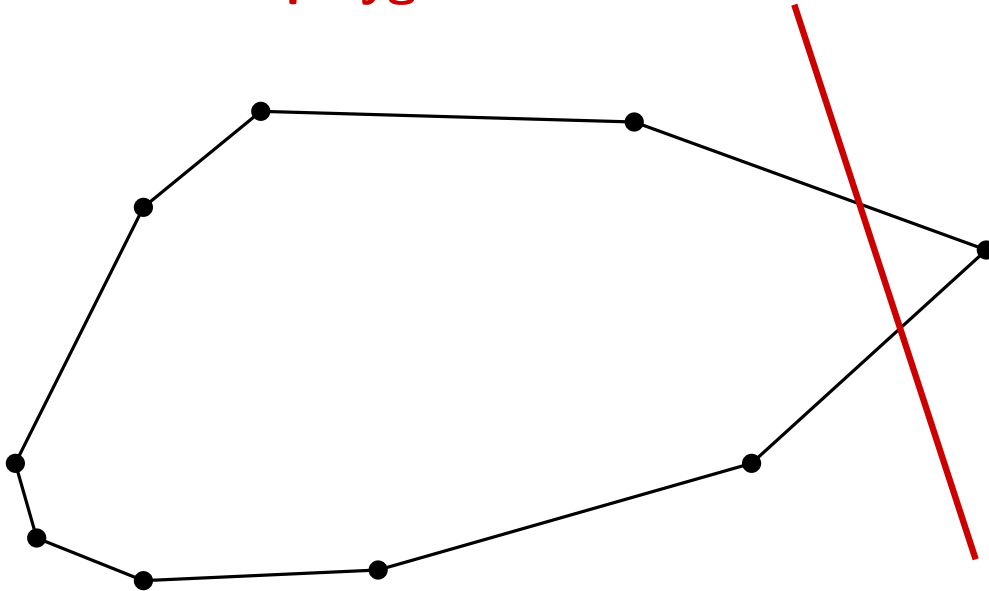
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Problem 3: Supporting lines point–polygon

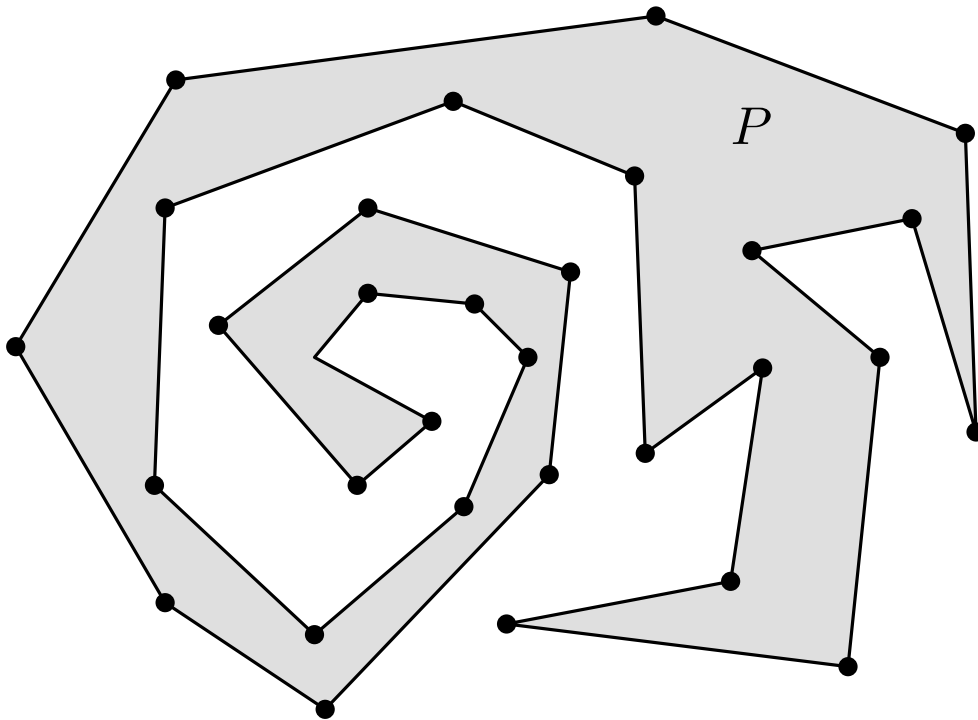
Input:

A polygon P with sorted vertices $p_1, p_2, \dots, p_n$

A point q not belonging to the convex hull of P

Output:

Lines through q and P that leave all of P to one side



SOME SIMPLE POLYGON PROBLEMS

Problem 3: Supporting lines point–polygon

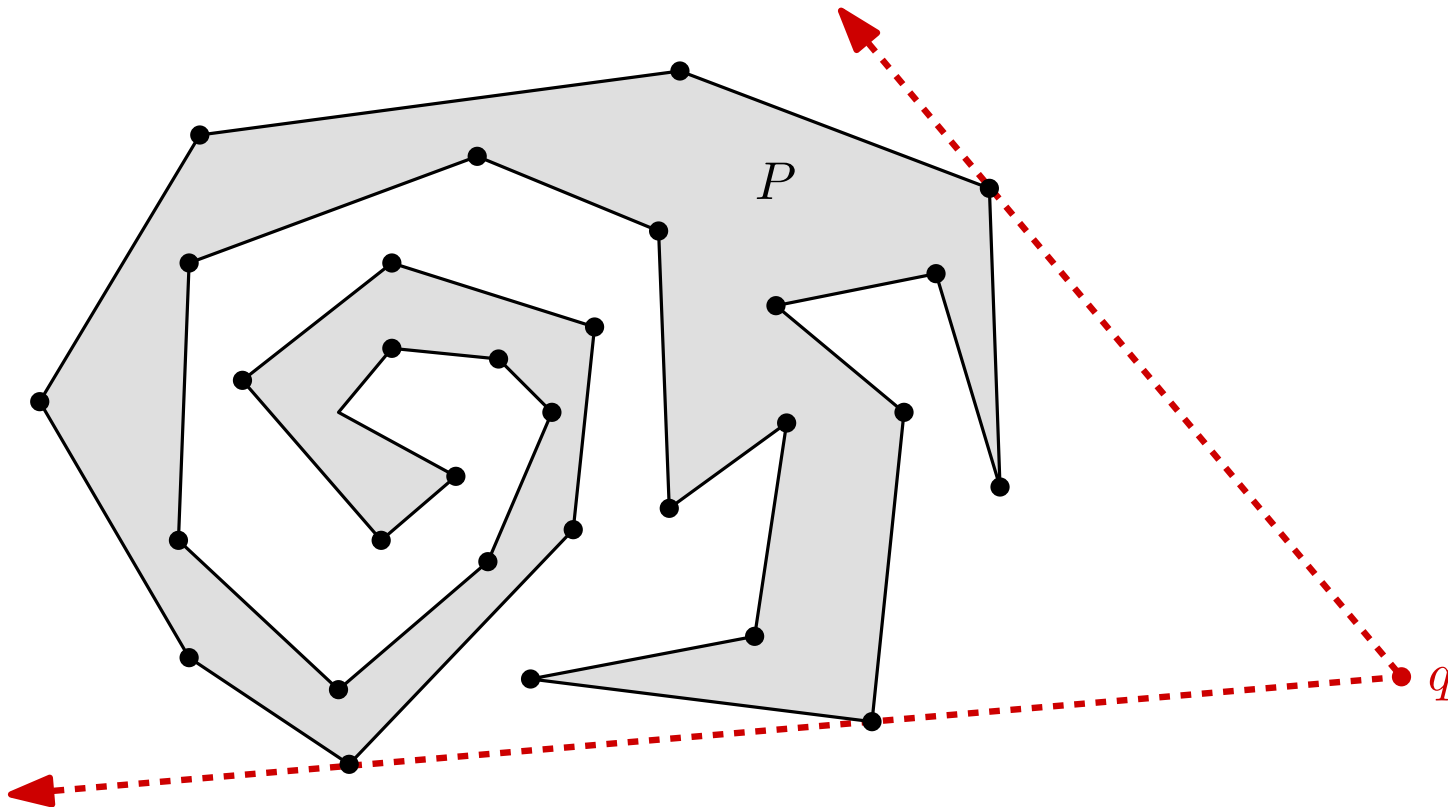
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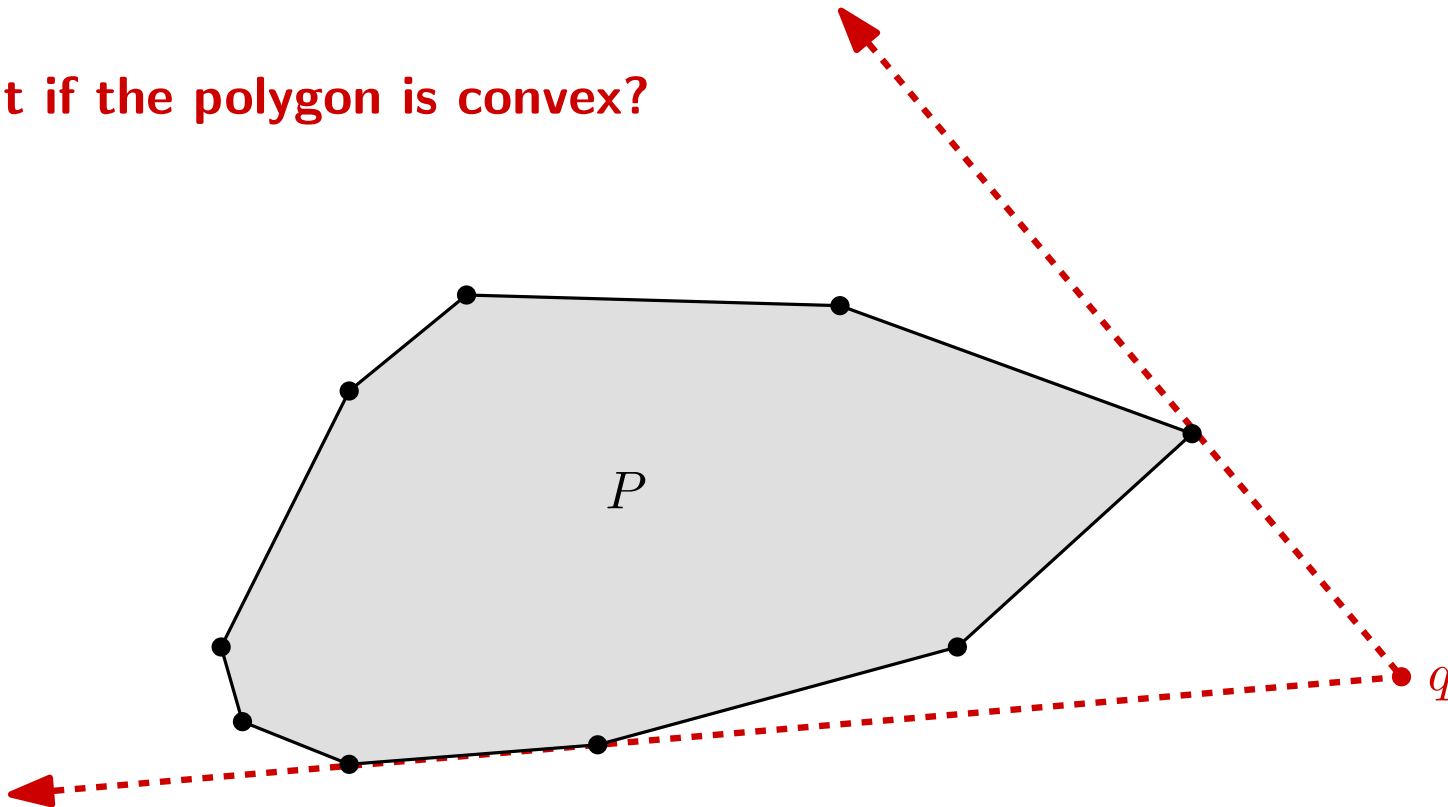
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What if the polygon is convex?

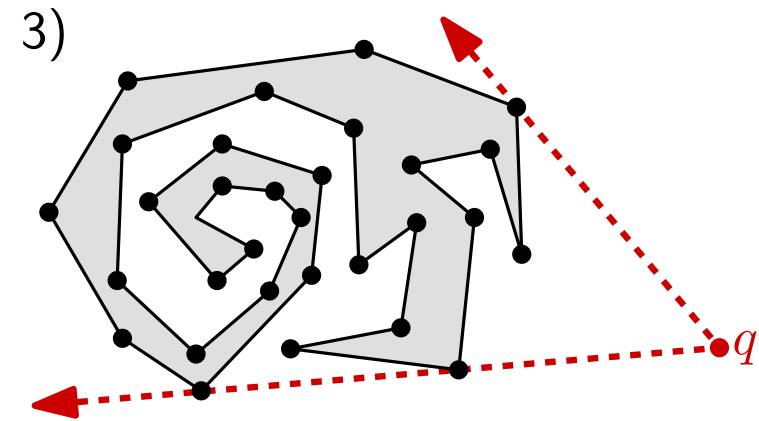
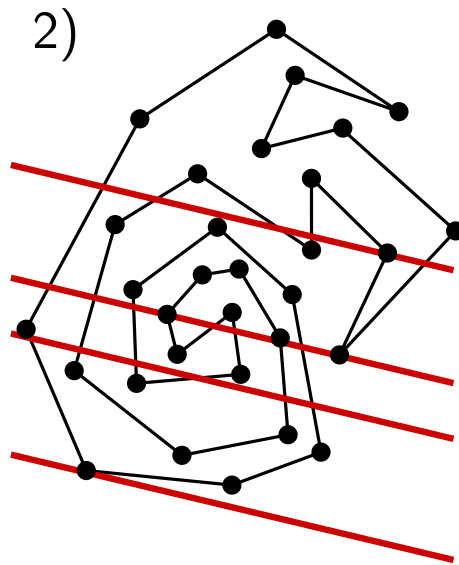
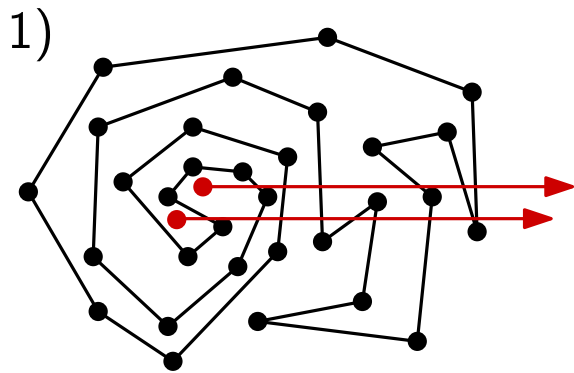


SOME SIMPLE POLYGON PROBLEMS

Key ideas of the solutions – general simple polygons

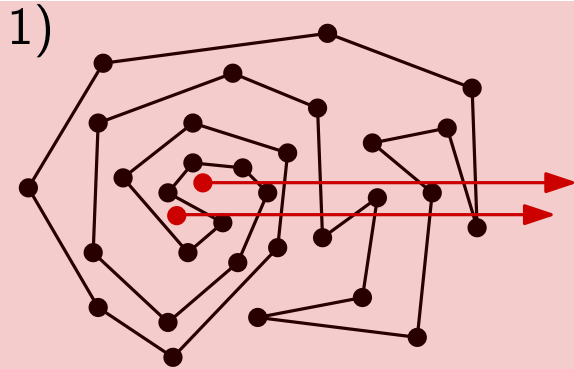
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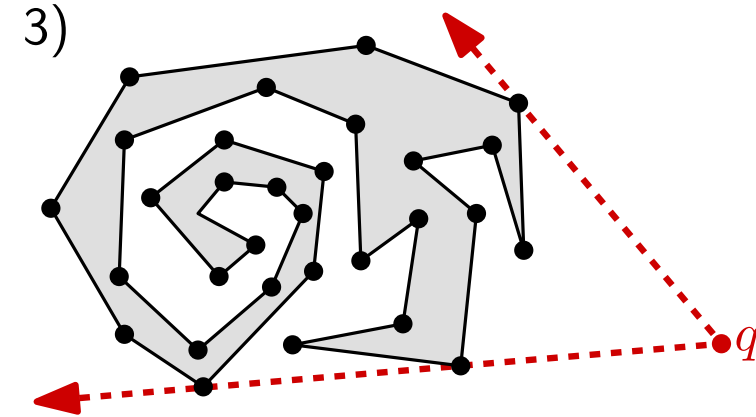
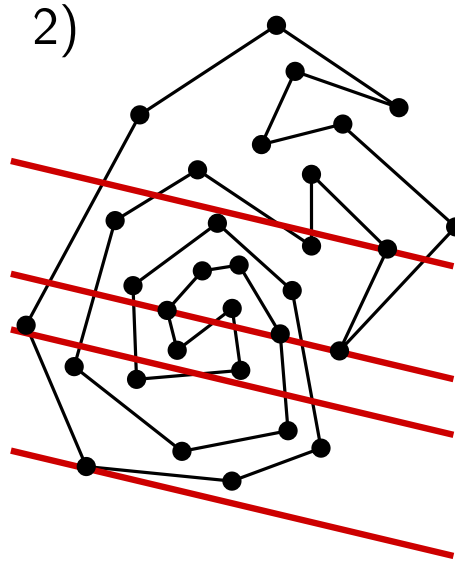
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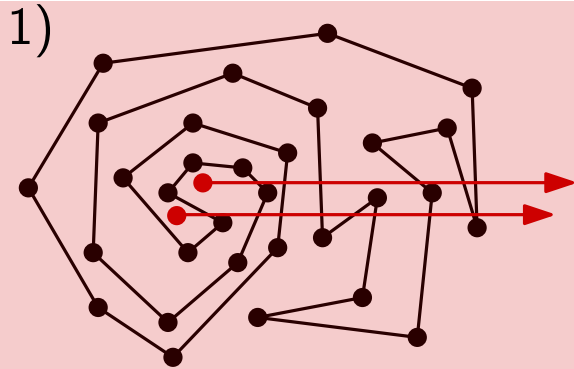
Geometric property:

$p \in P \Leftrightarrow$ The number of intersections of ∂P and any halfline with origin at p is odd.



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Key ideas of the solutions – general simple polygons



Geometric property:

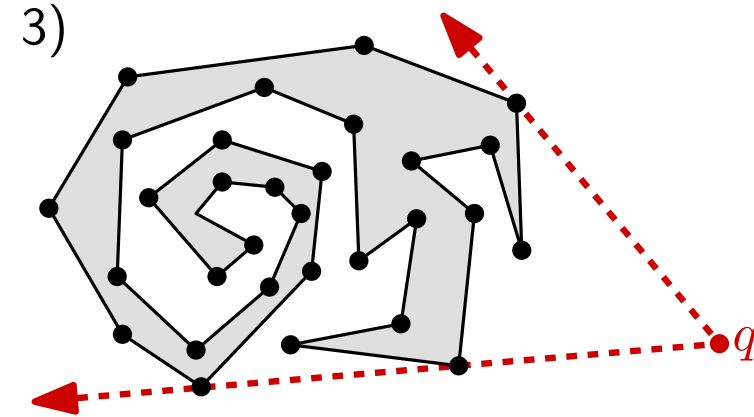
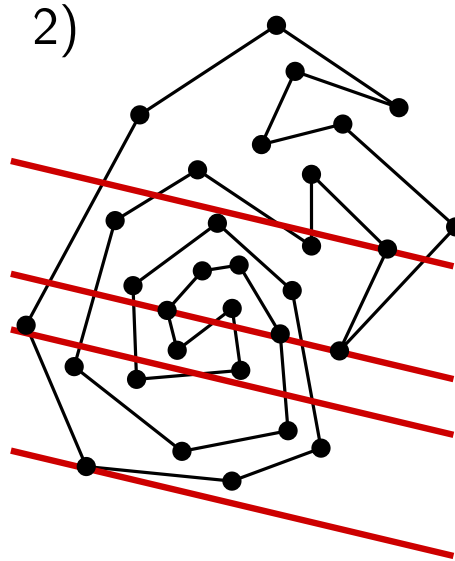
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Brute-force solution

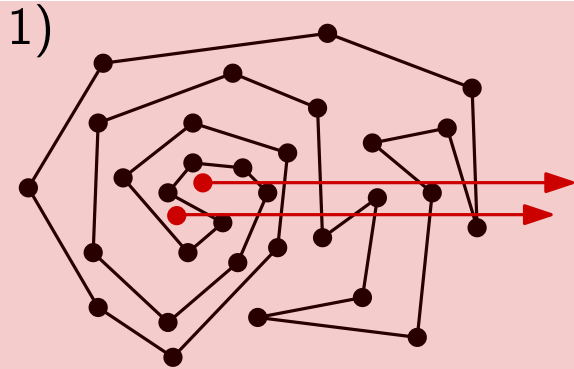
$O(n)$ time

$O(n)$ space



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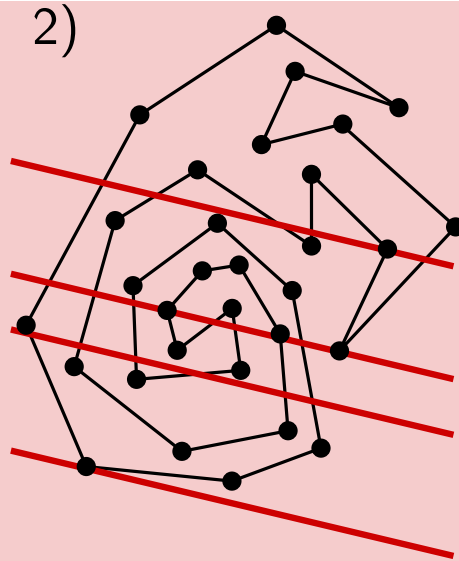
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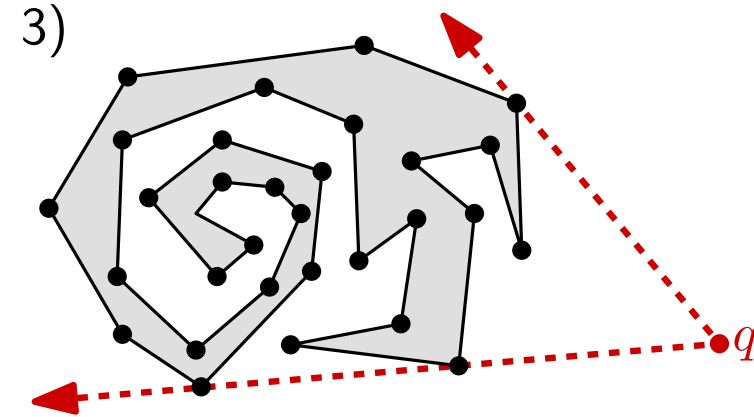
$O(n)$ time

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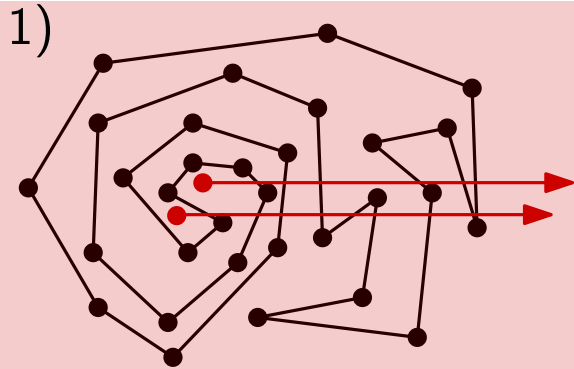
Geometric property:

No particular one.



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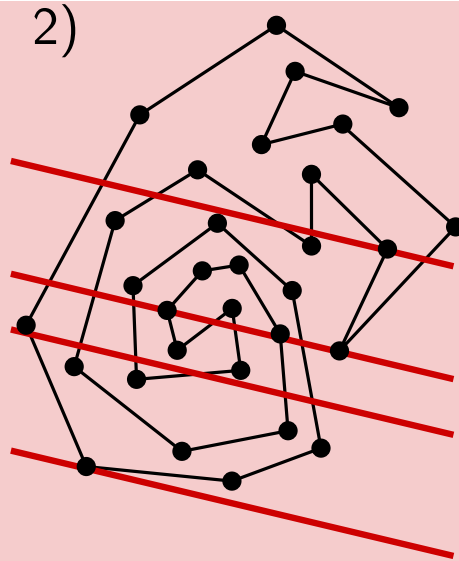
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Brute-force solution

$O(n)$ time
 $O(n)$ space



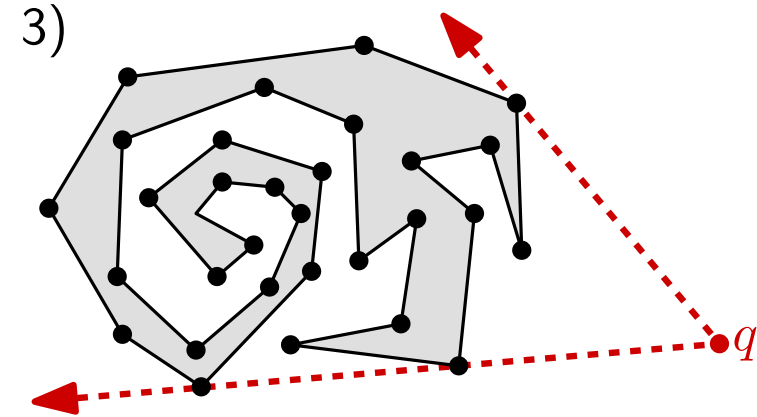
Geometric property:

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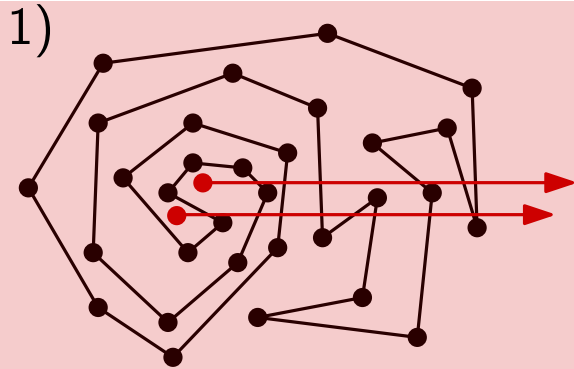
Brute-force solution

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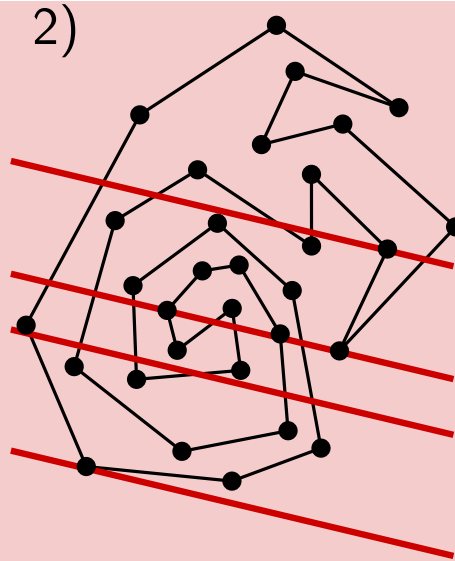
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Brute-force solution

$O(n)$ time
 $O(n)$ space



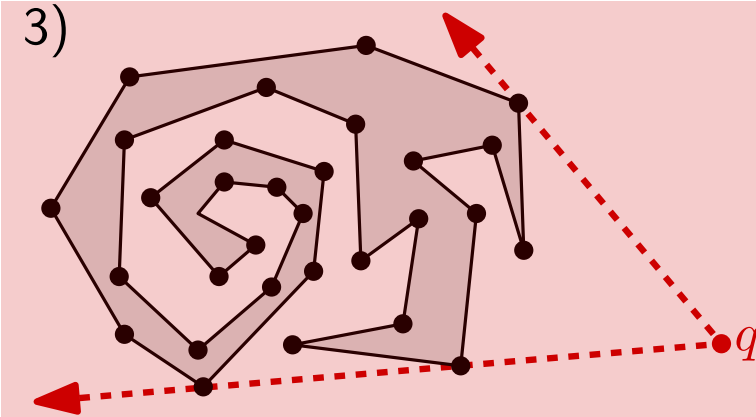
Geometric property:

No particular one.



Brute-force solution

$O(n)$ time
 $O(n)$ space

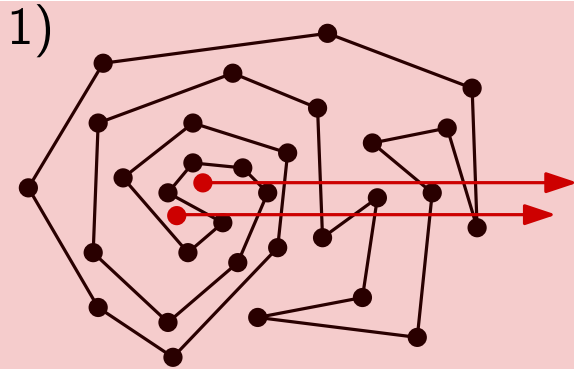


Geometric property:

The solutions are the angularly extreme vertices of P as seen from p .

SOME SIMPLE POLYGON PROBLEMS

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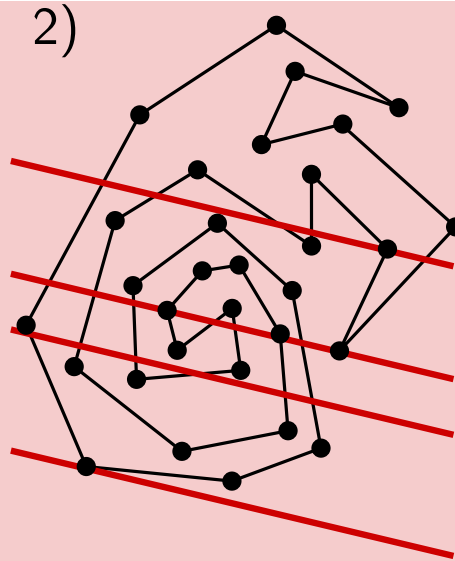
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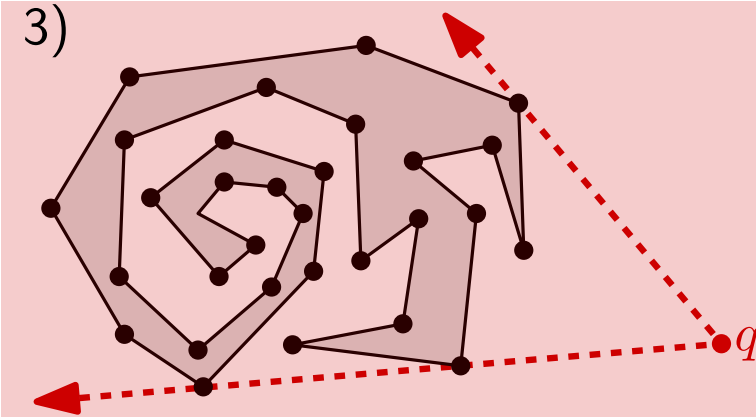
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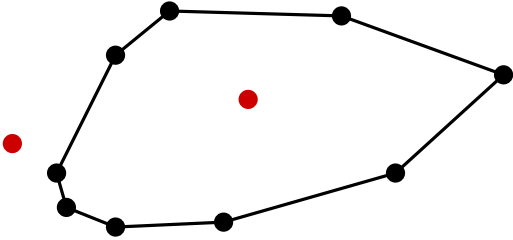
Use a max/min algorithm

$O(n)$ time
 $O(n)$ space

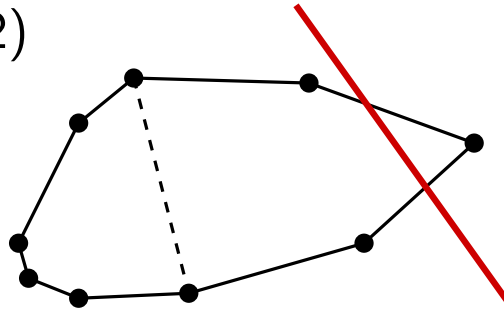
SOME SIMPLE POLYGON PROBLEMS

Key ideas of the solutions – convex polygons

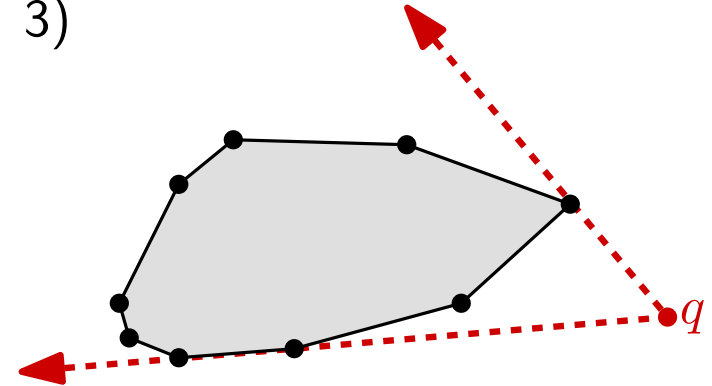
1)



2)



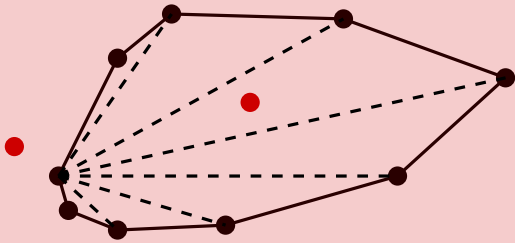
3)



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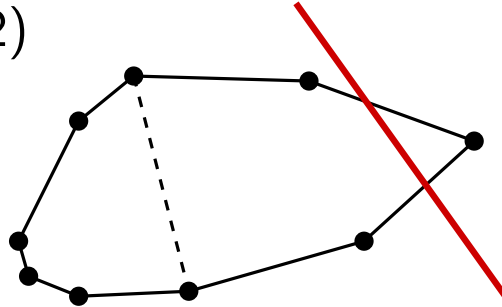
1)



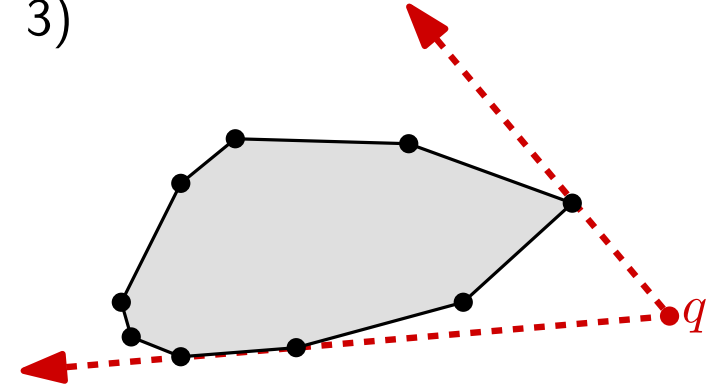
Geometric property:

We can decompose interior of polygon into sorted triangles, and binary search there

2)



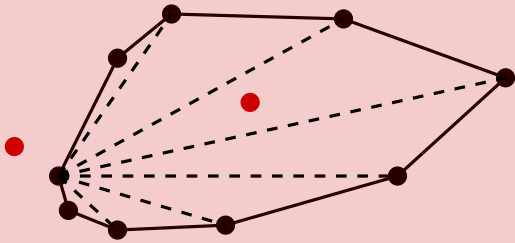
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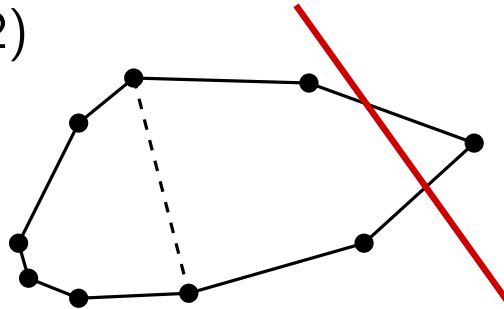
Binary search solution

$O(\log n)$ time

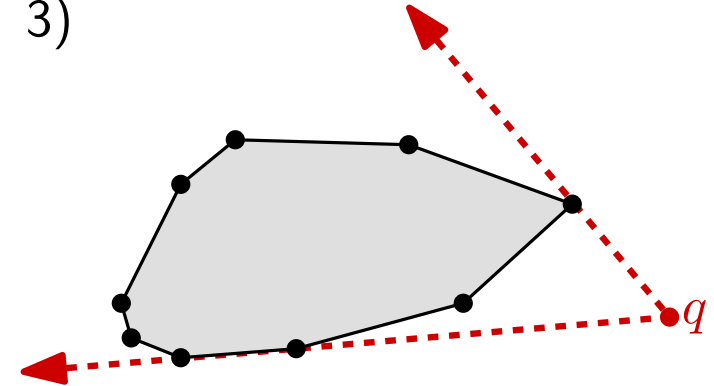
$O(n)$ space

(after preprocess)

2)



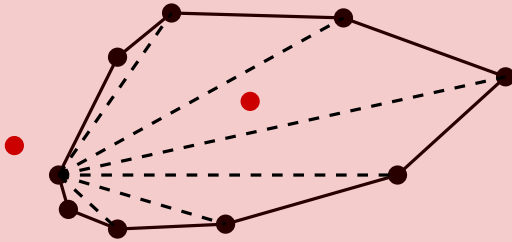
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Key ideas of the solutions – convex polygons

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Geometric property:

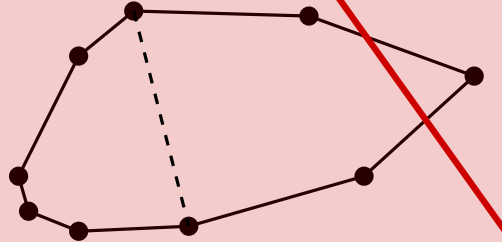
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Binary search solution

$O(\log n)$ time
 $O(n)$ space
(after preprocess)

2)



Geometric property:

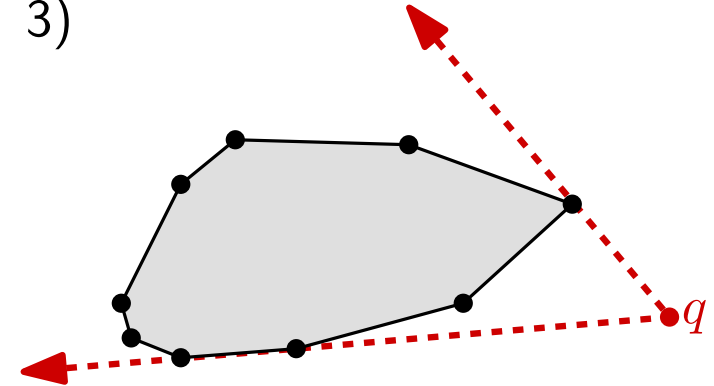
The “middle segment” allows us to recurse on only half of the polygon



Binary search solution

$O(\log n)$ time
 $O(n)$ space
(after preprocess)

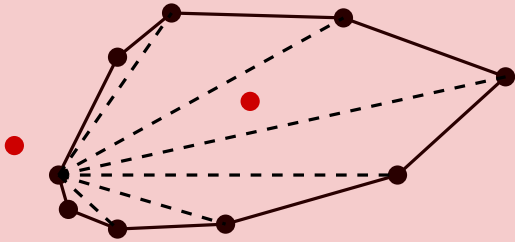
3)



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Key ideas of the solutions – convex polygons

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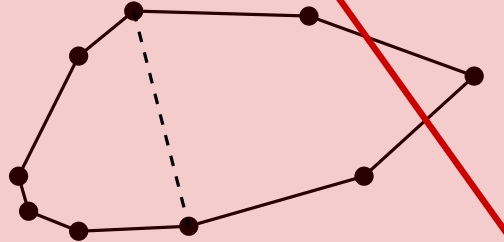
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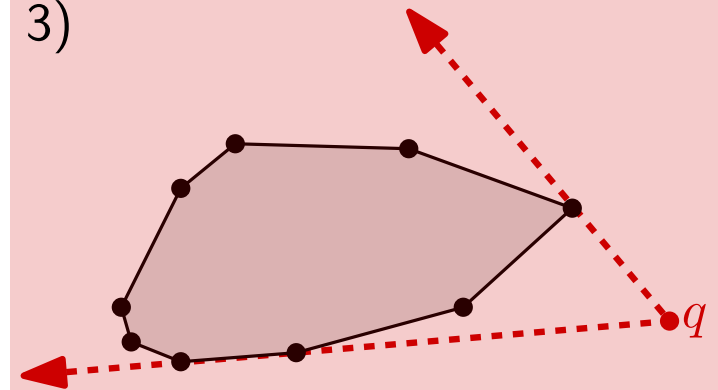
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Binary search solution

$O(\log n)$ time
 $O(n)$ space
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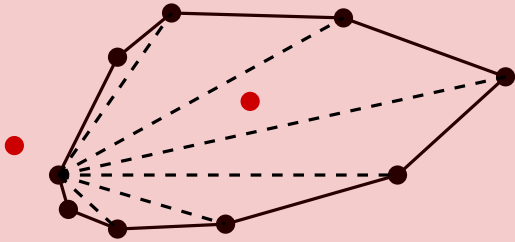
Geometric property:

Angle w.r.t. q is unimodal along ∂P .

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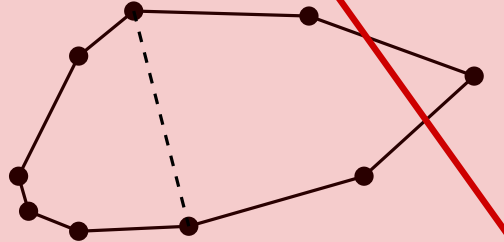
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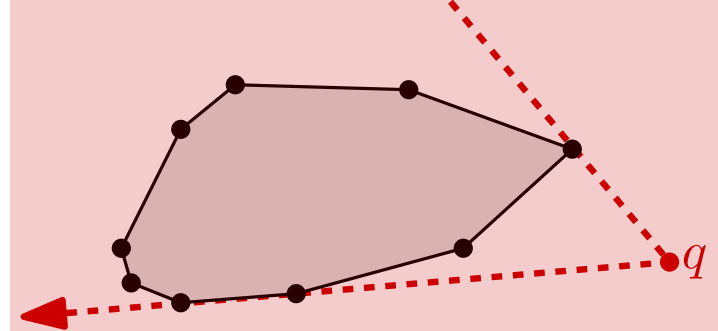
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Binary search solution

$O(\log n)$ time
 $O(n)$ space
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Binary search solution

$O(\log n)$ time
 $O(n)$ space
(after preprocess)

FURTHER READING

J. O'Rourke

Computational Geometry in C

Cambridge University Press, 1994 (2nd ed. 1998), pp. 17-35.

F. P. Preparata and M. I. Shamos

Computational Geometry: An Introduction

Springer-Verlag, 1985, pp. 36-45.